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Optimal Protein Intake Guide

Written by the editors of Examine.com

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Table of Contents

How much protein do you need per day?

As with most things in nutrition, there's no simple answer. Your ideal intake of calories and protein depends on your health, body composition, main goal, and the type, intensity, duration, and frequency of your physical activity. And even taking all this into account, you'll end up with a *starting* number, which you'll need to adjust through self-experimentation.

💡 Tip: Calculating your caloric needs

Your height, weight, age, and level of physical activity all contribute to your caloric needs. There are many calorie calculators out there, but the NIH Body Weight Planner stands out. It has been tested and validated against real-world data[1] and can estimate the number of calories *you* need to reach then maintain a specific weight.

Calorie-wise, there are only three types of diets:

- A **hypocaloric diet** feeds you fewer calories than you burn. If you want to lose weight, that's the diet for you. If you want most of your weight loss to be in the form of fat, not muscle, you'll also need to get enough protein and preferably to exercise.
- A **hypercaloric diet** feeds you more calories than you burn. If you want to gain weight, that's the diet for you. If you want most of your weight gain to be in the form of muscle, not fat, you'll need to get enough protein and engage in resistance training (by lifting weights, for instance).
- A **eucaloric diet** feeds you as many calories as you burn. It is also called a maintenance diet, since your weight won't change much; but you can gain or lose fat or muscle, depending on how much protein and exercise you get.

Daily protein requirements are expressed in grams, either per kilogram of body weight (g/kg) or per pound of body weight (g/lb). Ranges in the table below reflect known individual variances.

Optimal daily protein intake for adults (g/kg*)

<table>
<tr><th></th><th colspan="2">Of healthy weight</th><th>Overweight</th><th>Pregnant</th><th>Lactating</th></tr>
<tr><td>Sedentary</td><td colspan="2">1.2–1.8</td><td rowspan="2">1.2–1.5</td><td>≥1.8</td><td>≥1.5</td></tr>
<tr><td>Active</td><td>1.4–2.0</td><td>1.6–2.4</td><td colspan="2" rowspan="2">unknown</td></tr>
<tr><td>Goal</td><td>Maintenance</td><td>Muscle gain</td><td>Fat loss</td></tr>
</table>

Maintenance: eucaloric diet | **Muscle gain:** eucaloric diet (if sedentary) or hypercaloric diet (if active) | **Fat loss:** hypocaloric diet | * Grams per kilogram of body weight

- If you're sedentary, aim for **1.2–1.8 g/kg** (0.54–0.82 g/lb). Keep in mind that your body composition is more likely to improve if you add regular activity, especially resistance training, than if you merely hit a protein target.
- If you're of healthy weight, active, and wish to keep your weight, aim for **1.4–2.0 g/kg** (0.64–0.91 g/lb). People who are trying to keep the same weight but improve their body composition (more muscle, less fat) may benefit from the higher end of the range.
- If you're of healthy weight, active, and wish to build muscle, aim for **1.6–2.4 g/kg** (0.73–1.10 g/lb). Intakes as high as 3.3 g/kg may help experienced lifters minimize fat gain when bulking.
- If you're of healthy weight, active, and wish to lose fat, aim for **1.6–2.4 g/kg** (0.73–1.10 g/lb), skewing toward the higher end of this range as you become leaner or if you increase your caloric deficit (by eating less or exercising more). Intakes as high as 3.1 g/kg may enhance fat loss and minimize muscle loss in lean lifters.
- If you're overweight, aim for **1.2–1.5 g/kg** (0.54–0.68 g/lb). This range, like all the others in this list, is based on your total body weight (most studies on people who are overweight report their findings based on total body weight, but you'll find some calculators that determine your optimal protein intake based on your lean mass or your ideal body weight). If you're overweight, fat loss should be your priority, but that doesn't mean you cannot build some muscle over the same period. (Overweight includes obesity.)
- If you're pregnant, aim for **1.7–1.8 g/kg** (0.77–0.82 g/lb).
- If you're lactating, aim for at least **1.5 g/kg** (0.68 g/lb).
- If you're vegan or obtain most of your protein from plants, then your protein requirements may be higher because plant-based proteins are usually inferior to animal-based proteins with regard to both bioavailability and amino acid profile.

Also, note that …

- Protein intake should be based on body weight, not on caloric intake. (But caloric intake should be based on body weight, too, so the two intakes are linked.)
- Most studies have looked at dosages up to 1.5 g/kg; only a few have looked at dosages as high as 2.2–3.3 g/kg. However, in healthy people, even those higher dosages don't seem to have negative effects.

> How much protein you need depends on several factors, such as your weight, your goal (weight maintenance, muscle gain, or fat loss), your being physically active or not, and whether you're pregnant or not.

💡 Tip: Calculating your protein needs

You can quickly and easily calculate your optimal daily intake with our protein intake calculator.

Optimal daily protein intake for healthy, sedentary adults

For adults, the US Recommended Dietary Allowance (RDA) for protein is 0.8 g/kg.[2] However, a more appropriate statistical analysis of the data used to establish the RDA suggests this number should be higher: 1.0 g/kg.[3]

Note that, contrary to popular belief, the RDA doesn't represent an *ideal* intake. Instead, it represents the *minimum* intake needed to prevent malnutrition. Unfortunately, the RDA for protein was determined from nitrogen balance studies, which require that people eat experimental diets for weeks before measurements are taken. This provides ample time for the body to adapt to low protein intakes by down-regulating processes that are not necessary for survival but are necessary for optimal health, such as protein turnover and immune function.[4]

An alternative method for determining protein requirements, called the Indicator Amino Acid Oxidation (IAAO) technique, overcomes many of the shortcomings of nitrogen balance studies.[5] Notably, it allows for the assessment of protein requirements within 24 hours, thereby not leaving the body enough time to adapt. Studies using the IAAO method have suggested that about 1.2 g/kg is a more appropriate RDA for healthy young men,[6] older men,[7] and older women.[8][9]

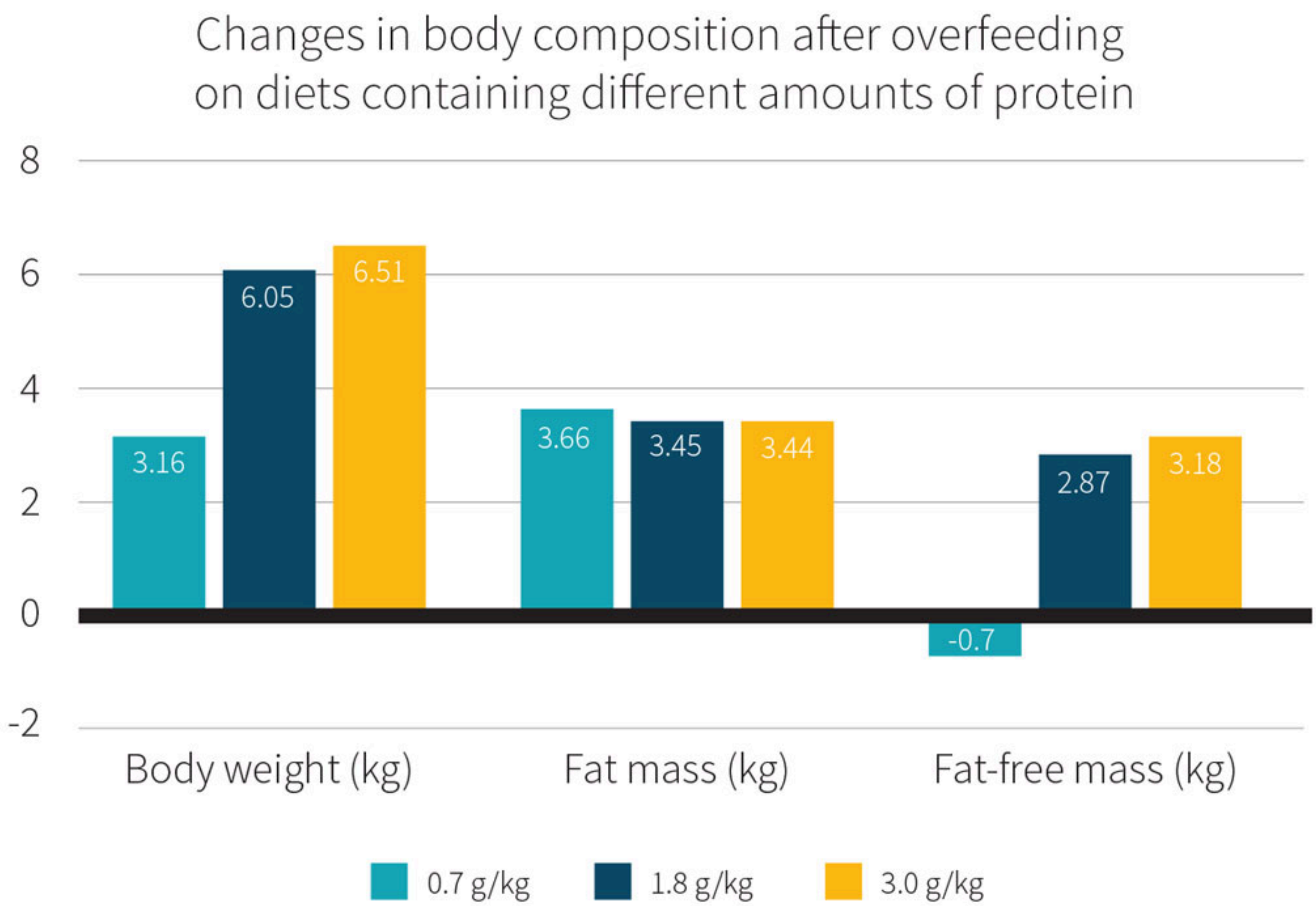
Further evidence that the current RDA for protein is not sufficient comes from a randomized controlled trial that confined healthy, sedentary adults to a metabolic ward for eight weeks.[10] The participants were randomized into three groups:

Three types of hypercaloric diets (40% above maintenance)

Macronutrients	Low protein	Normal protein	High protein
Protein (g/kg)	0.7	1.8	3.0
Protein (% of caloric intake)	5	15	25
Fat (% of caloric intake)	54	44	34

Macronutrients	Low protein	Normal protein	High protein
Carbohydrate (% of caloric intake)	41	41	41

Each diet was equally hypercaloric: each participant consumed 40% more calories than they needed to maintain their weight. Yet, as shown in the figure below, eating near the RDA for protein resulted in loss of lean mass, and while this loss is so small as to be nonsignificant, the higher protein intakes were associated with *increases* in lean mass.



Another takeaway from this study is that eating more than 1.8 g/kg doesn't seem to meaningfully benefit body composition, which makes it a good higher end for your daily protein intake, provided that you aren't physically active or trying to lose weight.

Optimal daily protein intake for healthy, sedentary adults

Body weight (lb)	Body weight (kg)	Lower end (g)	Higher end (g)
100	45	54	82
125	57	68	102
150	68	82	122
175	79	95	143
200	91	109	163
225	102	122	184
250	113	136	204
275	125	150	225
300	136	163	245

> The RDA for protein (0.8 g/kg) underestimates the needs of healthy, sedentary adults, who should rather aim for 1.2–1.8 g/kg (0.54–0.82 g/lb).

Optimal daily protein intake for athletes

If you're physically active regularly, you need more protein daily than if you were sedentary. The American College of Sports Medicine, the Academy of Nutrition and Dietetics, and the Dietitians of Canada recommend 1.2–2.0 g/kg to optimize recovery from training and to promote the growth and maintenance of lean mass when caloric intake is sufficient.[11] This recommendation is similar to that of the International Society of Sports Nutrition (ISSN): 1.4–2.0 g/kg.[12]

Importantly, it may be better to aim for the higher end of the above ranges. According to the most comprehensive meta-analysis to date on the effects of protein supplementation on muscle mass and strength, the average amount of protein required to maximize lean mass is about 1.6 g/kg, and some people need upwards of 2.2 g/kg.[13] Those of you interested in a comprehensive breakdown of this study will find one in *NERD* #34 (August 2017).

However, only 4 of the 49 included studies were conducted in people with resistance training experience (the other 45 were in newbies). IAAO studies in athletes found different numbers: on training days, female athletes required 1.4–1.7 g/kg;[14][15] the day following a regular training session, male endurance athletes required 2.1–2.7 g/kg;[16] two days after their last resistance-training session, amateur male bodybuilders required 1.7–2.2 g/kg.[17]

Since higher protein intakes seem to have no negative effects in healthy people, one may want to err toward the higher amounts. For most athletes (and similarly active adults), the ISSN range (1.4–2.0 g/kg) will work well:

Optimal daily protein intake for athletes

Body weight (lb)	Body weight (kg)	Lower end (g)	Higher end (g)
100	45	64	91
125	57	79	113
150	68	95	136
175	79	111	159

Body weight (lb)	Body weight (kg)	Lower end (g)	Higher end (g)
200	91	127	181
225	102	143	204
250	113	159	227
275	125	175	249
300	136	191	272

> Athletes and similarly active adults can optimize body composition, performance, and recovery with a daily protein intake of 1.4–2.0 g/kg (0.64–0.91 g/lb) and a preference toward the upper end of this range.

Optimal daily protein intake for muscle gain

Resistance training, such as lifting weights, is of course required for muscle gain: you can't just feed your muscles what they need to grow; you also need to give them a *reason* to grow.[18]

To gain muscle, most people should aim for 1.6[13]–2.4 g/kg.[19][20][21][22][23]

Assuming progressive resistance overload and a mildly hypercaloric diet (370–800 kcal above maintenance), a few studies suggest you'll gain less fat if you eat more protein (3.3 g/kg rather than 1.6–2.4 g/kg),[24][20] although one did not.[21]

What's important to understand is that a daily protein intake of 3.3 g/kg isn't likely to help you build more muscle than a daily protein intake of 1.6–2.4 g/kg. What the higher number can do is help you minimize the fat gains you'll most likely experience if you eat above maintenance in order to gain (muscle) weight.

Optimal daily protein intake for muscle gain

Body weight (lb)	Body weight (kg)	Lower end (g)	Higher end (g)
100	45	73	109
125	57	91	136
150	68	109	163
175	79	127	191
200	91	145	218
225	102	163	245
250	113	181	272

Body weight (lb)	Body weight (kg)	Lower end (g)	Higher end (g)
275	125	200	299
300	136	218	327

Hypercaloric diet. Intakes as high as 3.3 g/kg may help experienced lifters minimize fat gain while bulking.

> Athletes and active adults can optimize muscle gain with a daily protein intake of 1.6–2.4 g/kg (0.73–1.10 g/lb). For experienced lifters on a bulk, up to 3.3 g/kg (1.50 g/lb) may help minimize fat gain.

Optimal daily protein intake for fat loss

First, let it be clear that, though it is possible to lose fat on a **eucaloric diet** (aka maintenance diet — a diet that provides as many calories as you burn) by shifting your macronutrient ratios toward more protein, if you want to keep losing weight you'll need to switch to a **hypocaloric diet** (i.e., you'll need to start eating fewer calories than you burn).

High protein intakes help preserve lean mass in dieters, especially lean dieters. To optimize body composition, dieting athletes (i.e., athletes on a hypocaloric diet) should consume 1.6–2.4 g/kg,[25][26] skewing toward the higher end of this range as they become leaner or if they increase their caloric deficit (by eating less or exercising more).

Later studies have argued that, to minimize lean-mass loss, dieting lean resistance-trained athletes should consume 2.3–3.1 g/kg (closer to the higher end of the range as leanness and caloric deficit increase).[27] This latter recommendation has been upheld by the International Society of Sports Nutrition (ISSN)[28] and by a review article on bodybuilding contest preparation.[29]

Optimal daily protein intake for fat loss (if you're an athlete)

Body weight (lb)	Body weight (kg)	Lower end (g)	Higher end (g)
100	45	73	109
125	57	91	136
150	68	109	163
175	79	127	191
200	91	145	218
225	102	163	245
250	113	181	272

Body weight (lb)	Body weight (kg)	Lower end (g)	Higher end (g)
275	125	200	299
300	136	218	327

Hypocaloric diet. Intakes as high as 3.1 g/kg may enhance fat loss and minimize muscle loss in lean lifters.

Note that those recommendations are for people who are relatively lean already and trying to lose a little more fat while preserving their precious muscle mass. Several meta-analyses involving people with overweightness or obesity suggest that 1.2–1.5 g/kg is an appropriate daily protein intake range to maximize fat loss.[30][31][32] This range is supported by the European Association for the Study of Obesity, which recommends up to 1.5 g/kg for elderly adults with obesity.[33] It is important to realize that this range is based on actual body weight, not on lean mass or ideal body weight.

🔍 Digging Deeper: Why more protein for athletes?

Dieting athletes benefit from higher protein intakes, relative to their weights, than overweight and obese dieters. This can be partly explained in three interrelated ways:

- Overweight and obese individuals have sluggish metabolisms that tend to favor fat storage over protein storage (protein being stored as muscle). They usually need a greater caloric deficit to lose fat than do athletes (this rule isn't absolute, since the closer an athlete gets to essential body fat, the harder it gets to lose fat).
- Protein intake is based on total weight. Let's consider two dieters who weigh the same. If one is an already lean athlete and the other an overweight individual, the latter will get to consume a lot fewer calories. Let's say I'm an athlete consuming 3,000 kcal and 180 grams of protein (720 kcal): my diet is 24% protein. Now let's say I'm overweight and consuming 2,000 kcal and 120 grams of protein (480 kcal): my diet is also 24% protein.
- If an already lean athlete and an overweight person weigh the same, the former has more muscle and so needs more protein to maintain muscle mass.

Considering the health risks associated with overweightness and obesity,[34][35] it is also noteworthy that eating a diet higher in protein (27% vs. 18% of calories) significantly reduces several cardiometabolic risk factors, including waist circumference, blood pressure, and triglycerides, while also increasing satiety.[36] These effects are small, however, and likely dependent on the amount of body fat one loses.

Optimal daily protein intake for fat loss (if you're overweight)

Body weight (lb)	Body weight (kg)	Lower end (g)	Higher end (g)
100	45	54	68
125	57	68	85
150	68	82	102
175	79	95	119
200	91	109	136
225	102	122	153
250	113	136	170
275	125	150	187
300	136	163	204

Hypocaloric diet. If you're overweight or obese, fat loss should be your priority, but that doesn't mean you cannot build some muscle over the same period.

> When dieting for fat loss, athletes and other active adults who are already lean may maximize fat loss and muscle retention with a daily protein intake of 1.6–2.4 g/kg (0.73–1.10 g/lb). People who

are overweight or obese are best served by consuming 1.2–1.5 g/kg (0.54–0.68 g/lb).

Quickly and easily calculate your optimal daily intake with our protein intake calculator.

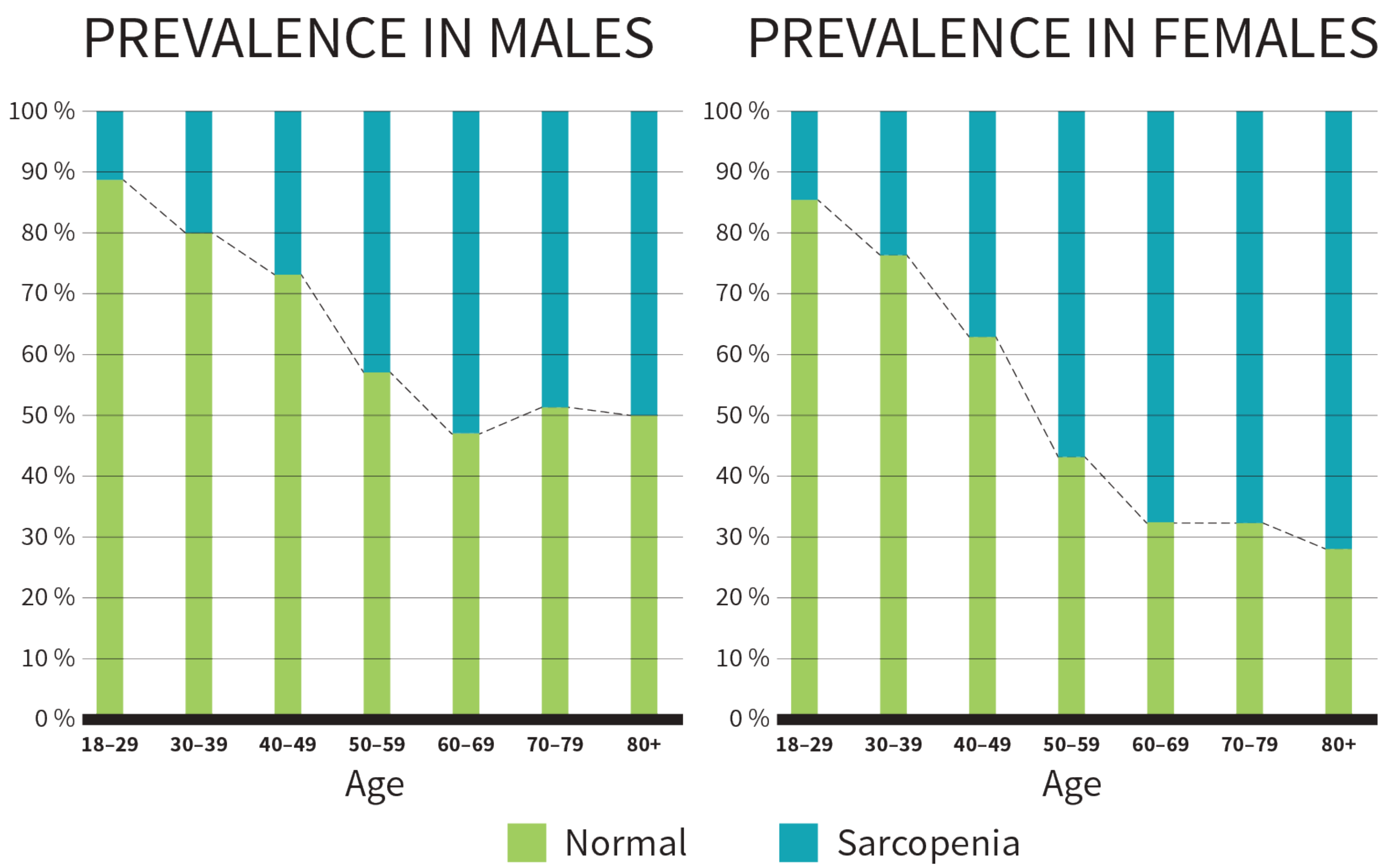
Optimal daily protein intake for older adults

Sarcopenia is a muscle disorder. It is defined as an impairment of physical function (walking speed or grip strength) combined with a loss of muscle mass.[37][38] It is the primary age-related cause of frailty.

Frailty[39] is associated with a higher risk of disabilities that affect your ability to perform daily activities,[40] a higher risk of having to go to a nursing home,[41] and a higher risk of experiencing falls,[42] fractures,[43] and hospitalizations.[44]

The link between sarcopenia, frailty, and associated morbidities may explain why sarcopenia is associated with a greater risk of premature death and reduced quality of life.[45][46] This isn't a rare issue, either: in the US, over 40% of men and nearly 60% of women over the age of 50 have sarcopenia, and more than 10% of people in their 20s.[47]

Prevalence of sarcopenia by age and sex in the US



Reference: Janssen et al. *J Am Geriatr Soc*. 2002.[47]

Fortunately, sarcopenia is neither inevitable nor irreversible — some seniors have built more muscle in their old age than they ever had in their youth. The older you get, though, the greater your muscles' anabolic resistance (i.e., their resistance to growth),[48] and so the greater the protein intake and exercise volume (of resistance training, preferably) you'll need to stimulate muscle protein synthesis.[49][50][51]

The protein RDA for adults over 50 is currently the same as for younger adults: 0.8 g/kg.[2] Same as for younger adults, however, studies using the IAAO method have suggested that 1.2 g/kg would be a more appropriate RDA.[7][8][9] Moreover, since a low protein intake is associated with frailty and worse physical function than a higher protein intake,[52][53] several authorities now recommend that older adults consume 1.2–1.5 g/kg.[54][55][56] Finally, while all adults have similar daily protein requirements,[57] older adults have higher per-meal requirements.

Notably, doubling protein intake from 0.8 to 1.6 g/kg has been shown to significantly increase lean body mass in elderly men.[58] Similar observations have been made in elderly women who increase their protein intake from 0.9 to 1.4 g/kg.[59] Even a small increase in protein intake from 1.0 to 1.3 g/kg has minor benefits towards lean mass and overall body composition.[60]

So how much protein should *you* get?

- Sedentary but healthy seniors: 1.0–1.2 g/kg (0.45–0.54 g/lb)
- Sick or injured seniors: 1.2–1.5 g/kg (0.54–0.68 g/lb)
- Seniors wishing to lose weight: 1.5–2.2 g/kg (0.68–1.00 g/lb)
- Seniors wishing to build muscle: 1.7–2.0 g/kg (0.77–0.91 g/lb)

🔍 Digging Deeper: Conflicting data?

You might have noticed that, according to the bulleted list above, **sedentary, healthy seniors** (1.0–1.2 g/kg) need less protein than other **sedentary, healthy adults** (1.2–1.8 g/kg).

Does that make sense?

No, frankly, it doesn't. Of course, not every senior suffers from sarcopenia and not every young adult is free of it, as shown in the graphic above, but the *range* for healthy, sedentary seniors should not be lower than the *range* for other healthy, sedentary adults.

This glaring discrepancy springs from our having to rely on different sets of studies — thus different data sets — for seniors and for other adults. The data sets appear to conflict because they're both incomplete, both imperfect (if they weren't, we wouldn't need more studies).

In each section of this article, we decided to stick to the most relevant data available, but if you're a healthy, sedentary senior, you can also decide to adopt the higher range we gave for other healthy, sedentary adults: 1.2–1.8 g/kg.

Daily protein intake based on *body weight* (BW)

BW	BW	0.36	0.45	0.45	0.68	0.77	0.91	1.00	g/lb
lb	**kg**	**0.8**	**1.0**	**1.2**	**1.5**	**1.7**	**2.0**	**2.2**	**g/kg**
100	45	36	45	54	68	77	91	100	g

BW	BW	0.36	0.45	0.45	0.68	0.77	0.91	1.00	g/lb
lb	**kg**	**0.8**	**1.0**	**1.2**	**1.5**	**1.7**	**2.0**	**2.2**	**g/kg**
125	57	45	57	68	85	96	113	125	g
150	68	54	68	82	102	116	136	150	g
175	79	64	79	95	119	135	159	175	g
200	91	73	91	109	136	154	181	200	g
225	102	82	102	122	153	173	204	225	g
250	113	91	113	136	170	193	227	250	g
275	125	100	125	150	187	212	249	275	g
300	136	109	136	163	204	231	272	299	g

Depending on their health statuses and goals, older adults (50+ years) should aim for a daily protein intake of 1.0–2.2 g/kg (0.45–1.00 g/lb).

Optimal daily protein intake for pregnant women

The protein RDA for pregnant women is 1.1 g/kg.[2] This value was estimated by adding three values:

- The RDA for a healthy adult (0.8 g/kg)
- The amount of additional body protein a pregnant woman accumulates
- The amount of protein used by the developing fetus

However, as we saw previously with non-pregnant healthy adults, the RDA may not be sufficient, let alone optimal. There's some IAAO evidence that the RDA for pregnant women should be about 1.66 g/kg during early gestation (weeks 11–20) and 1.77 g/kg during late gestation (weeks 32–38).[61][62] Moreover, a meta-analysis of 16 intervention studies reported that protein supplementation during pregnancy led to reduced risks for the baby:[63]

- 34% lower risk of low gestational weight
- 32% lower risk of low birth weight
- 38% lower risk of stillbirth

This effect was more pronounced in undernourished women than in adequately nourished women. Importantly, these values were determined from sedentary women carrying one child, meaning that pregnant women who engage in regular physical activity or are supporting the growth of more than one child may need even higher amounts.

Also, keep in mind that we can only tell you what the studies reported; we can't possibly know about *your* health and *your* pregnancy specifically. Please be sure to consult with your *obstetrician/gynecologist* (ob/gyn) before making any changes.

Optimal daily protein intake for pregnant women

Body weight (lb)	Body weight (kg)	Weeks 11–20 (g)	Weeks 32–38 (g)
100	45	≥75	≥80

Body weight (lb)	Body weight (kg)	Weeks 11–20 (g)	Weeks 32–38 (g)
125	57	≥94	≥100
150	68	≥113	≥120
175	79	≥132	≥141
200	91	≥151	≥161
225	102	≥169	≥181
250	113	≥188	≥201
275	125	≥207	≥221
300	136	≥226	≥241

Pregnant women may require a daily protein intake of at least 1.77 g/kg (0.8 g/lb) to support both the fetus and themselves. Protein supplementation during pregnancy appears to lower some risks for the baby — including the risk of stillbirth — especially in undernourished women.

Optimal daily protein intake for lactating women

As with pregnancy, there is little research investigating how lactation and breastfeeding affect protein requirements.[64] Women produce a wide range of breast milk volumes, regardless of their energy status (i.e., milk production is maintained even among underweight women — i.e., women with a BMI under 18.5).[65] The infant's demands appear to be the primary regulator of milk production.[66][67]

Based simply on adult protein requirements plus the protein output in breast milk, the RDA for lactating women was set at 1.3 g/kg.[2] However, one study reported that half of the lactating women consuming 1.5 g/kg were in negative nitrogen balance,[68] while another study suggested that 1.0–1.5 g/kg leads to a rapid downregulation of protein turnover suggestive of an adaptive response to insufficient intake.[69]

Considering (1) the lack of data on the effects of a protein intake greater than 1.5 g/kg in lactating women and (2) that consuming 1.5 g/kg or less leads to adaptations suggestive of insufficient intake, lactating women should aim to consume *at least* 1.5 g/kg of protein daily.

Optimal daily protein intake for lactating women

Body weight (lb)	Body weight (kg)	Protein intake (g)
100	45	≥68
125	57	≥85
150	68	≥102
175	79	≥119
200	91	≥136
225	102	≥153

Body weight (lb)	Body weight (kg)	Protein intake (g)
250	113	≥170
275	125	≥187
300	136	≥204

Lactating women should aim for a daily protein intake of *at least* 1.5 g/kg.

Quickly and easily calculate your optimal daily intake with our protein intake calculator.

Optimal daily protein intake for infants and children

Optimal daily protein intake for infants and children in grams per kilogram of body weight (g/kg)

	Infants (preterm)	Infants (0–6 months)	Infants (7–12 months)	Toddlers (1–3 years)	Children (4–13 years)
Sedentary	3.0–4.0	≥1.5	≈3.0	3.0–4.0	≥1.5 g/kg
Active	n/a	n/a	n/a	n/a	unknown

Optimal daily protein intake for infants

Healthy infants

The *adequate* protein intake of healthy infants **aged 0–6 months**, based on their average weight and milk intake, is 1.52 g/kg.[70]

The *average* protein intake of healthy infants **aged 7–12 months** is estimated at 1.6 g/kg,[71] assuming that half their protein comes from breast milk and half from complementary foods. Yet the RDA is set at 1.2 g/kg for this age group based entirely on studies conducted in toddlers and children.[72]

Preterm infants

Preterm infants need to be fed enough protein to promote growth rates similar to those observed in healthy fetuses growing *in utero*. The following daily intakes have been recommended based on gestational age:[73]

- 3.5–4.0 g/kg (less than 30 weeks)
- 2.5–3.5 g/kg (30–36 weeks)

- 2.5 g/kg (more than 36 weeks)

Moreover, a systematic review by the Cochrane Collaboration reported greater weight gain and higher nitrogen accretion in preterm infants whose protein intake was 3.0–4.0 g/kg, compared to lower daily intakes.[74] These findings were echoed by another systematic review of 24 clinical trials.[75]

Since breast milk doesn't contain enough protein to meet these requirements, complementary supplementation is standard practice.[76][77]

Formulas

Breast milk is considered the optimal source of nutrition for infants (0–12 months old) and is recommended as the exclusive source of nutrition for non-preterm infants aged 0–6 months. However, not all infants can breastfeed. Infant formulas provide an alternative, but there are considerable differences in composition from breast milk.[78] One such difference is the protein content, which tends to be higher in formula.

Compared to exclusive breastfeeding, formula feeding is associated with greater increases in fat-free mass throughout the first year of life. Fat mass and body fat percentage tend to be lower during the first six months, but play catch-up afterward and ultimately end up higher with formula feeding than with breastfeeding.[79]

An association was found between formula feeding, faster growth during infancy, and obesity in childhood, adolescence, and young adulthood.[80] Some researchers suggested that the higher protein content of infant formulas was responsible,[81] but others have argued that there are too many contributing factors (e.g., breastfeeding helps infants learn to better regulate their energy intake) to single one out.[82]

Moreover, if the higher protein content of formulas were responsible for the infants' accelerated growth, then how could we explain the similar growth of infants fed formulas containing 1.2 or 1.7 grams of protein per 100 milliliters,[83] or formulas containing 1.0, 1.3, or 1.5 grams of protein per 100 milliliters?[84] (For reference, breast milk contains about 1 gram of protein per 100 milliliters.)

Still, even if consuming more protein from formulas than would be obtained from breast milk is not necessarily detrimental, it doesn't appear to confer a benefit. There is no good reason to stray from the nutrient composition of mother's milk during infancy, unless dealing with a preterm infant.

Meat

When complementary foods are introduced to infants during the latter half of infancy (7–12 months), there may be a benefit to more protein from meat.[85] Compared to feeding cereal grains alongside breast milk (total protein: 1.4 g/kg/day), feeding pureed meats alongside breast milk (total protein: 2.9 g/kg/day) was shown to lead to better growth without excess fat gain.[86]

Another study demonstrated that, as a complementary food, meat led to more favorable growth patterns than dairy (higher length-for-age and lower weight-for-length) by 12 months of age[87] — differences that persisted at the age of 2 years.[88] Both the meat group and the dairy group consumed the same total protein (3.0 g/kg).

> During their first six months, healthy infants should consume at least 1.5 grams of protein per kilogram of body weight per day (≥1.5 g/kg/day). This intake can be achieved exclusively through breastfeeding. From age 6 to 12 months, they should consume around 3.0 g/kg/day (and could especially benefit from using meat as complementary food). Preterm infants require 3.0–4.0 g/kg/day to facilitate catch-up growth.

Optimal daily protein intake for toddlers

The same data used to establish the RDA for infants aged 7–12 months (1.2 g/kg) was used to determine the RDA for toddlers aged 1–3 years (1.05 g/kg).[2] The average daily protein intake of US toddlers is 4.0 g/kg, with 90% of US toddlers consuming over 3.0 g/kg.[89]

There is a dearth of data for this age group. However, in toddlers aged 2 years with a total daily protein intake of 4.0 g/kg, complementary protein from meat led to better growth (higher length-for-age) than the same amount of complementary protein from dairy.[88]

> There is little research on what is *optimal*, but the *average* daily protein intake of US toddlers is 4 g/kg — nearly four times the RDA. Meat appears to be a better complementary food than milk.

Optimal daily protein intake for children

The protein RDA is slightly higher for children (4–13 years) than for adults: 0.95 versus 0.8 g/kg.[2] This difference makes sense considering that children are still growing and need more protein to facilitate the process. As with adults, however, the RDA may underestimate true requirements.

Use of the IAAO technique in children aged 6–11 years has suggested that around 1.5 g/kg would make for a more appropriate RDA.[90] Protein requirements are likely higher in children involved in sports and other athletic activities.[91]

There are no long-term studies on optimal protein intake since it would be unethical to deprive children of the protein they need for their development and various physiologic and metabolic functions.

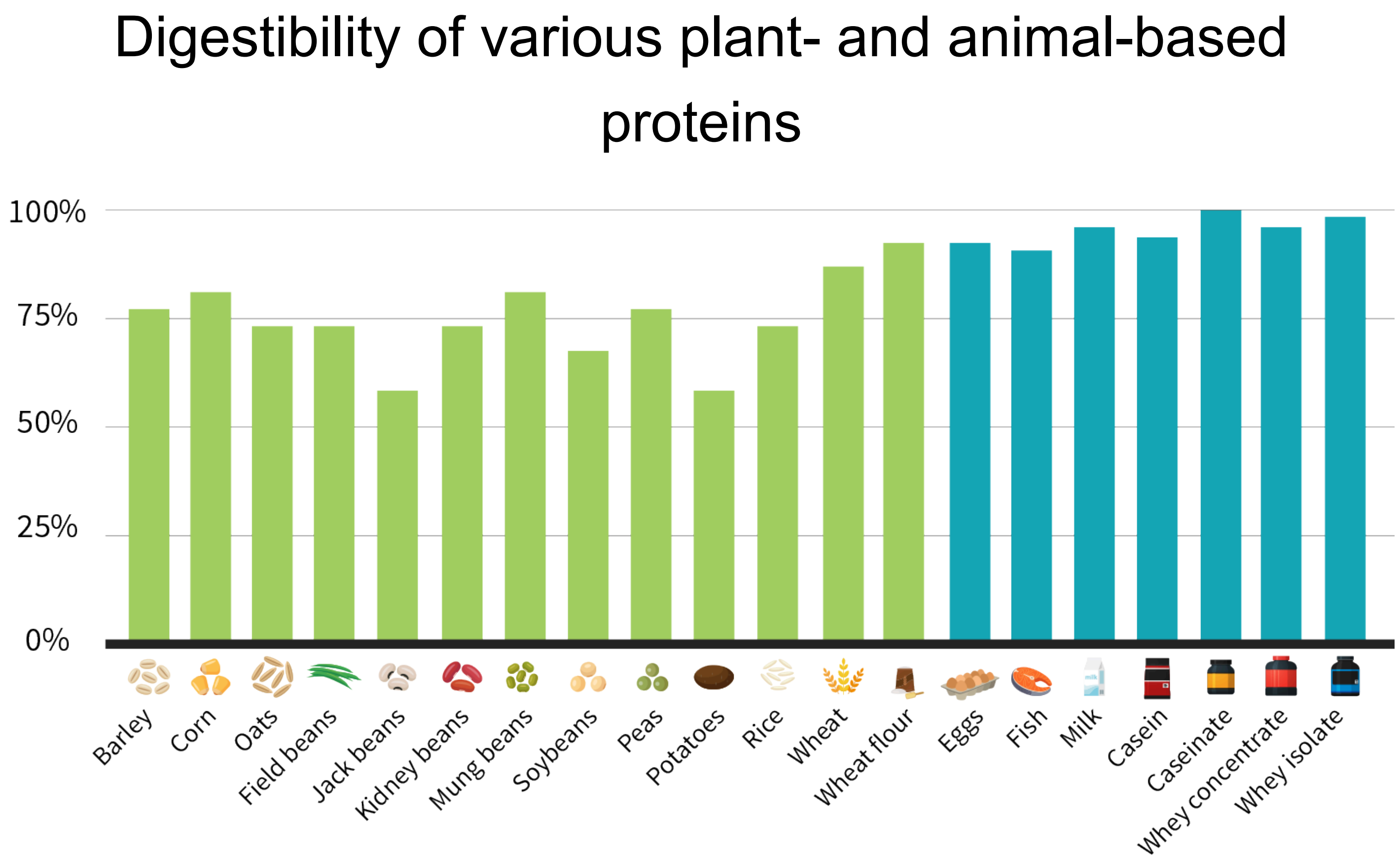
> Children require at least 1.5 grams of protein per kilogram of body weight per day (1.5 g/kg/day). An unknown amount of additional protein is likely required by children who are involved in sports or otherwise regularly active.

Optimal daily protein intake for vegetarians and vegans

The protein requirements discussed so far were based on studies that used animal-based protein supplements, such as whey or egg protein supplements, or were conducted mostly in omnivores. There is no reason to believe that people who get their protein mostly or entirely from plants have inherently different protein requirements, but since plant-based proteins tend to be lower in quality than animal-based proteins, if you obtain most of your protein from plants you will need to pay attention not just to the amount of protein you eat but also to the quality of that protein.[92]

A protein's quality is determined by its **digestibility** and **amino acid profile**.

Digestibility matters because if you don't digest and absorb some of the protein you eat, then it may as well not have been eaten. Animal-based proteins consistently demonstrate a digestibility rate higher than 90%, whereas proteins from the best plant-based sources (legumes and grains) show a digestibility rate of 60–80%.[93]



Reference: FAO. *Protein Quality Evaluation in Human Nutrition*. 2013.[94]

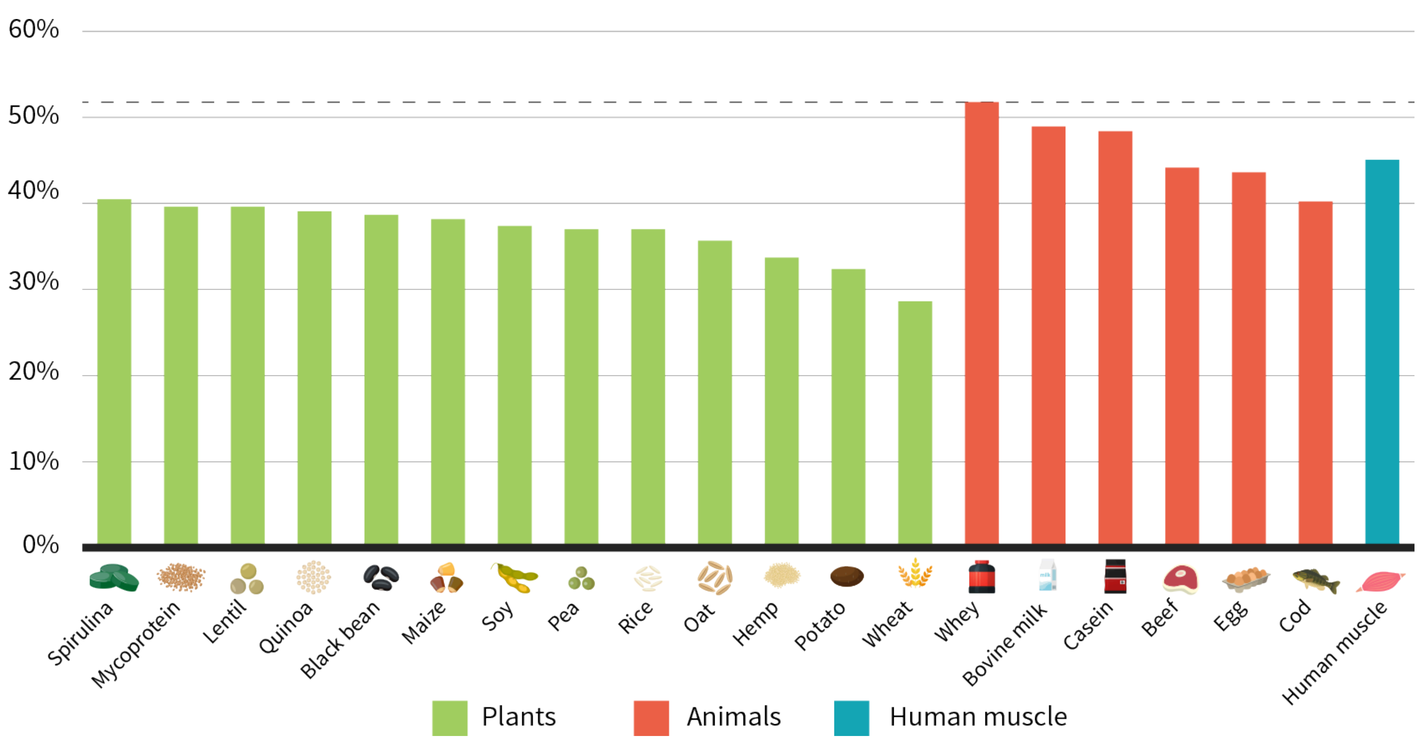
Plants contain anti-nutrients that inhibit protein digestion and absorption, such as trypsin inhibitors, phytates, and tannins.[95] While cooking does reduce anti-nutrient concentrations, it doesn't eliminate them entirely. Plant-based protein powders, however, are mostly free of antinutrients and so have digestibility rates similar to those of animal-based proteins.[93]

The **amino acid profile** of a protein matters because all proteins, including the protein you eat and the protein in your body, are made from some combination of 20 *amino acids* (AAs). Your body can produce 11 of these AAs, making them *nonessential amino acids* (NEAAs). Your body cannot produce the other 9, which are therefore *essential amino acids* (EAAs) you must get through food.

Building muscle requires that, cumulatively, *muscle protein synthesis* (MPS) exceeds *muscle protein breakdown* (MPB), resulting in a net accumulation of muscle protein. All 20 AAs are required to build muscle tissue,[96] but MPS is stimulated primarily by the EAAs in the food you ingest.[97]

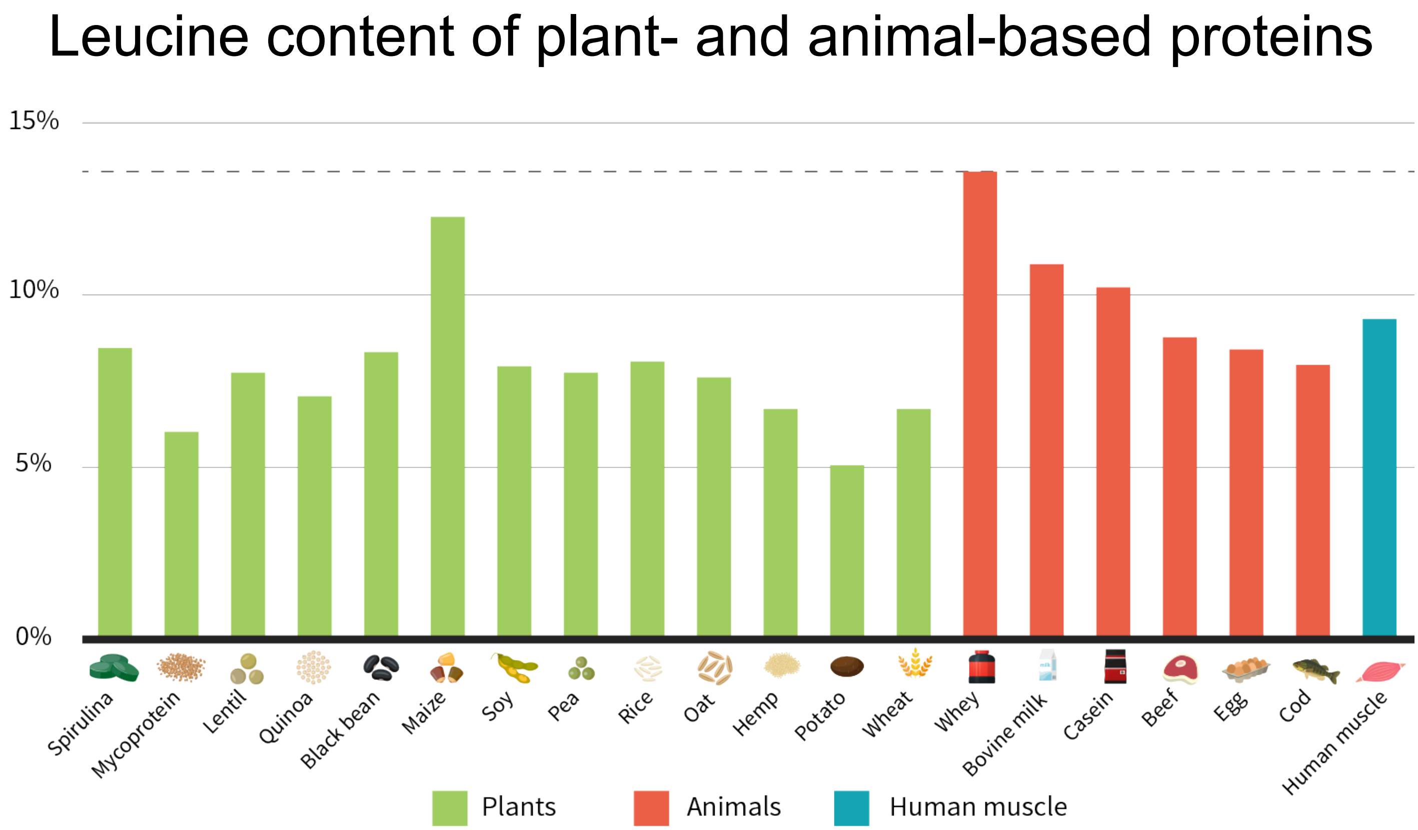
Plant-based proteins, whether from whole foods or protein powders, contain less EAAs than animal-based proteins.

EAA content of plant- and animal-based proteins



Reference: FAO. *Protein Quality Evaluation in Human Nutrition*. 2013.[94]

In particular, plant-based proteins are lower in the EAA leucine, which is believed to act as a signal to “turn on” anabolic signaling pathways and MPS,[98][99] although all EAAs are required for the effect to persist.[100]



Reference: van Vliet et al. *J Nutr*. 2015.[101]

The lower leucine and EAA content of plant-based proteins helps explain why several studies have reported lower rates of MPS from soy protein powders and beverages than from whey protein,[102][103] skim milk,[104] whole milk with cheese,[105] and lean beef.[106]

However, while differences in MPS do appear to translate to differences in lean mass when modest supplemental protein doses are used (about 20 g/day),[107][108] when higher doses are used (33–50 g/day), animal-based (whey) and plant-based (soy, rice) supplemental proteins appear to affect lean mass similarly.[109][110][111][112] In short, consuming more protein overall appears to offset the lower quality of the plant-based proteins.

Plant-based proteins also contain *limiting amino acids*, which are EAAs present in such small amounts that they bottleneck protein synthesis. Lysine is the most common limiting amino acid, especially in cereal grains, such as wheat and rice.[113] Nuts and seeds also tend to have lysine as a limiting amino acid. Beans and legumes, on the other hand, contain sufficient lysine but lack sulfurous amino acids, such as methionine and cysteine. Combining different plant-based proteins can help make up for their respective deficits.

> Plant-based proteins are of lower quality (they are less bioavailable and contain less EAAs). If you get most of your protein from plants, you will need to consume more protein to achieve the same muscle growth as someone with a more omnivorous diet.

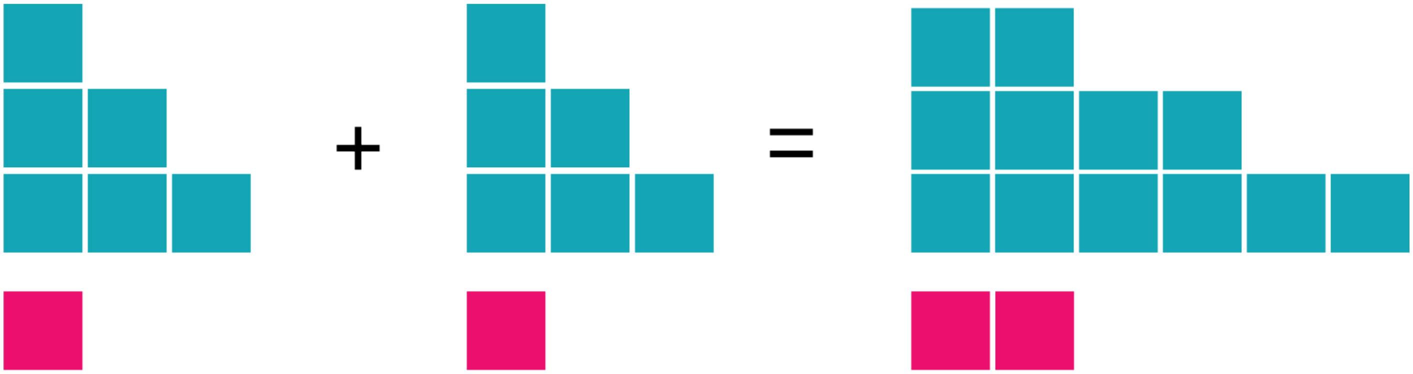
Bolstering plant-based proteins

The simplest method to overcome the EAA deficits of a plant protein is to eat more of it. As aforementioned, a handful of studies have shown that large doses (33–50 g/day) of animal-based (whey) and plant-based (soy, rice) supplemental proteins appear to increase lean mass similarly. [109][110][111][112]

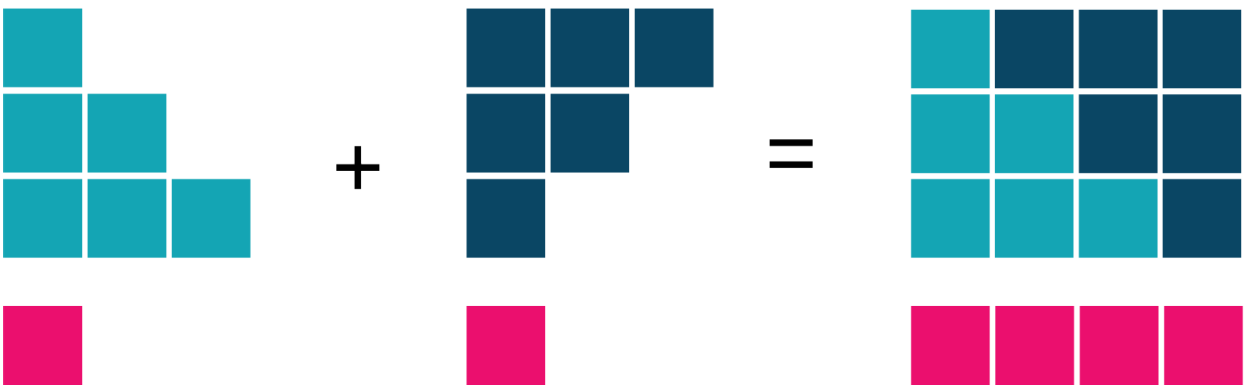
Another way to overcome the EAA deficits of plant proteins is to combine complementary EAA profiles.[114] Historic examples of such combinations include beans with corn in the Americas and rice with soybean in Asia. These grain-legume combos work because legumes supply the lysine missing in grains whereas grains supply the methionine and cysteine missing in legumes.

Combining incomplete proteins

Eating more of the same plant-based protein = not optimal



Combining complementary plant-based proteins = optimal



Adapted from: Woolf et al. *PLoS One*. 2011.[114]

Unfortunately, most plant proteins are low in leucine, meaning that combining different plant proteins will not have a large benefit unless one of those proteins is corn protein (whose leucine content rivals that of whey protein).

If your protein has less leucine, you need to eat more of it to maximize MPS — or you can take leucine as a supplement. MPS was increased similarly by 25 grams of whey protein (providing 3 grams of leucine) and by a combination of 6.25 grams of whey protein and 4.25 grams of supplemental leucine (5 grams of leucine in total).[115] A rodent study using plant proteins reported similar results.[116]

> The EAA deficits of plant-based proteins can be overcome by eating more, combining complementary proteins, and

supplementing with leucine.

How much protein per meal?

Muscle protein synthesis (MPS) is the process of building new skeletal muscle tissue. When MPS chronically exceeds *muscle protein breakdown* (MPB), resulting in a positive net protein balance, we can expect muscle growth over the long term.[117][118] Each time you eat represents an opportunity to promote muscle growth through the stimulation of MPS.

Protein-feeding studies using various doses of whey protein suggest that 0.24 g/kg/meal will maximize the MPS of the **average** young adult,[51] whereas 0.40 g/kg/meal will maximize the MPS of **most** young adults.[119] For older adults, these two values jump to 0.40 and 0.60 g/kg/meal.[51]

Desirable minimal protein intake range per meal and age

Body weight (lb)	Body weight (kg)	20s	30s, 40s, 50s	≥60
100	45	11–18	*13–24*	18–27
125	57	14–23	*16–30*	23–34
150	68	16–27	*20–36*	27–41
175	79	19–32	*23–42*	32–48
200	91	22–36	*26–48*	36–54
225	102	24–41	*30–54*	41–61
250	113	27–45	*33–60*	45–68
275	125	30–50	*36–66*	50–75
300	136	33–54	*39–72*	54–82

The ranges in this table represent individual variations. The minimum protein requirements increase as you age, but to what degree is uncertain because of the age gap left by the studies: most subjects were in their 20s (0.24–0.40 g/kg) or 60s/70s (0.40–0.60 g/kg). For people in their 30s, 40s, or 50s, the 0.29–0.53 g/kg range reflected in this table is an educated guess. **References:** Schoenfeld and Aragon. *J Int Soc Sports Nutr*. 2018. [120] Rafii et al. *J Nutr*. 2016. [7] Morton et al. *Front Physiol*. 2015. [119] Moore et al. *J Gerontol A Biol Sci Med Sci*. 2015. [51] Rafii et al. *J Nutr*. 2015. [8]

Your mileage may vary. The ranges above are not *ideal* ranges. Instead, they cover the known extent of interindividual variations among healthy adults. In other words, if you're in your 20s, you don't need to calculate your protein intake per meal so that it falls precisely within the 0.40–0.60 g/kg range. Rather, your *minimum* protein intake per meal (to maximize MPS) is likely to fall within that range.

Further, there are at least **three good arguments** in favor of eating toward or even above the higher end of your range:

First, the ranges we listed are derived from studies using whey protein in isolation. Whey protein is highly bioavailable, rich in *essential amino acids* (EAAs), and quickly digested. When eating lower-quality or slower-digesting proteins (as would occur when eating a meal, especially one rich in plant-based foods), higher protein intakes are probably required.

Second, while these values suggest a protein-intake threshold for maximally stimulating MPS, there is no known threshold for *whole-body* protein balance.[121] For example, a study using meals with lean beef found that 40 and 70 grams of protein (0.5 and 0.8 g/kg) led to similar increases in MPS, but that 70 grams led to greater increases in whole-body protein synthesis and greater decreases in whole-body protein breakdown.[122] In other words, eating more protein may not necessarily translate to greater *muscle*-protein turnover and growth, but since muscle tissue accounts for only 25–30% of whole-body protein turnover,[123] the additional protein is not "wasted" (a common myth).

Third, as shown above in the Prevalence of sarcopenia by age and sex in the US graphic, even people in their twenties can suffer from sarcopenia — in which case they would benefit from a protein intake closer to the one recommended in this table for adults over sixty.

You may have heard that if you eat more than 30 grams of protein in one sitting, the "excess" will pass undigested, but that's just a myth. It is however true that spreading your protein intake over a few meals, making sure that you meet your desirable minimal protein intake per meal with each meal, will generally result in greater lean mass and strength. A pragmatic review article suggests that, to maximize their lean mass, active adults should consume 1.6–2.2 g/kg/day spread across four meals (0.40–0.55 g/kg/meal).[120]

For maximal stimulation of muscle protein synthesis, aim for a per-meal dose of quality protein (such as can be found in meat, eggs, and dairy) of 0.4–0.6 g/kg. Higher doses will not be wasted and are probably necessary when eating mixed meals that contain a variety of protein sources.

🔍 Digging Deeper: Protein intake ceiling

You may have heard that you can't digest more than 30 grams of protein in one sitting. This notion of a "protein intake ceiling" derives partly from early studies that observed increased nitrogen losses in the urine with increased protein intakes. This was thought to mean that the extra protein was wasted.[124]

We now know that things aren't so simple. When you eat protein, your body doesn't use it directly; instead, it breaks it down into its constituent amino acids and uses those to make its own proteins. When you eat more protein, your body can afford to replace more of its damaged or oxidized proteins, so that your protein synthesis *and* breakdown are both increased.

In other words, eating more protein increases your body's protein turnover.[125] The raised levels of urinary nitrogen then reflect, not a waste of eaten protein, but an increase in the breakdown of your body's damaged or oxidized proteins.[126]

(Note that elevated levels of urinary nitrogen can also indicate health issues, such as problems with kidney function.)

The notion of a "protein intake ceiling" derives also from studies on the body's *muscle protein synthesis* (MPS) response to different intakes of protein.

- One study in healthy young men found that eating more than 20 grams of whole-egg protein didn't further increase MPS.[127]
- Another study in younger and older people found that 90 grams of protein from 90% lean beef didn't increase MPS more than did 30 grams.[128]

However, your body doesn't use dietary protein only to make muscle, or even only to make other proteins. It also uses the nitrogen from the dietary protein’s amino acids to synthesize important non-protein molecules, such as purines and pyrimidines, the building blocks for nucleic acids such as DNA and RNA.

Moreover, your small intestines are able to absorb and store a large amount of amino acids, ready to be used when your body needs them.

In short, the idea that eating more than 30 grams of protein results in wasted protein is incorrect. Your body will break down and use all the protein you eat, sooner or later, one way or another.

(Note, also, that higher protein intakes increase satiety, which is particularly helpful if you’re trying to cut calories as part of a weight-loss diet.)

How much protein after exercise?

After exercising, when your muscles are more sensitive to the anabolic effect of protein, take a dose in the range of your desirable minimal protein intake per meal. If you've been exercising on an empty stomach, you'll be in negative protein balance, so take this dose as soon as possible. Otherwise, try to take it within a couple of hours — the exact size of your "window of opportunity" depends on how much protein you're still digesting.[129]

🔍 Digging Deeper: After better than before?

Is it better to eat before or after exercising?

In a 2020 crossover trial we first summarized for Examine Personalized, 8 healthy young men were divided into three groups.[130]

All three groups performed a full-body strength workout in the morning. One group ate a meal 1.5 hours before, another just after, and the last waited for lunchtime.

This happened three times, with three days in between. Each time, the men switched to a different group. Thus, at the end of the trial, they had all tried all three meal times.

Blood insulin and certain modified amino acids were measured as indicators of muscle breakdown. **The results?** Eating a mixed-nutrient meal immediately after resistance training, rather than 1.5 hours before or not until lunchtime, resulted in the greatest suppression of muscle protein breakdown.

Whether or not this correlates with improvements in muscle growth and recovery, however, is unclear.

How to get enough protein

You've used our table or calculator to determine how much protein you need in a day, but the numbers don't look right. Let's say you're 125 pounds, of healthy weight, physically active, and trying to get even leaner. You discover that your optimal intake starts at **102 grams** of protein. Isn't that *too much* for someone so light?

So it may seem at first blush. But let's take a step back. Let's say you're 125 pounds, of healthy weight, sedentary, and just trying to keep the same body composition. Your optimal intake starts at **68 grams** of protein — so **272 kcal** (less than 16% of the maintenance daily calories of a sedentary 40-year-old, 125-pound, 5'4" female). Not so daunting, now, is it?

Next, you decide to add physical exercise, in order to get even leaner. If you are 125 pounds and run at 7.5 mph (8 minutes/mile) for just ½ hour, you burn 375 kcal, compared to 41 for computer work. In other words, you burn 334 kcal more than when sitting and typing — just about the least physically demanding activity.

If you took those **added kilocalories** solely as protein, that would make 84 grams of protein. Add 84 grams to your optimal protein intake when you *don't* exercise, and you get **152 grams** of protein — way more than your **102 grams** starting target. (Since protein isn't the best source of energy, you could instead choose to get 102 grams of protein and 50 grams of carbs and/or fat.)

We can also calculate from the other direction. You're 125 pounds and of healthy weight, going from sedentary to active in order to get even leaner: how will your protein intake change?

At least 102 grams of protein (active)

−

At least 68 grams of protein (sedentary)

=

At least 34 grams of additional protein

=

At least 136 additional kcal

In other words, to make **optimal** use of protein to lose fat *and preserve muscle* when you're 125 pounds and already of healthy weight, you need to exercise so as to burn, on average, for just **136 Calories** of extra protein. If you run at 7.5 mph (8 minutes/mile) for just ½ hour and take **34 grams** of extra protein but don't otherwise eat more than when you were sedentary and your body composition was stable, you'll end the day on a deficit of **198 kcal**.

Even a small caloric deficit will lead to weight loss, though your body actually plays by more complicated mathematics than the ones we've just used. In practice, you might want to increase your caloric deficit a little, either by reducing your intake of carbs and/or fat or simply by exercising a little more.

One last thing: how can you get **34 grams** of extra protein without much extra carbs or fat? You can either take **one heaping scoop** of protein powder, probably at the end of your workout, or modify your diet so as to eat more protein (and less fat and/or carbs) over the whole day, by including more protein-rich foods in your meals.

References

1. ^ Hall KD, et al. Quantification of the effect of energy imbalance on bodyweight. *Lancet*. (2011)
2. ^ a b c d e f Institute of Medicine. 10 Protein and Amino Acids. *Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids*. (2005)
3. ^ Elango R, et al. Evidence that protein requirements have been significantly underestimated. *Curr Opin Clin Nutr Metab Care*. (2010)
4. ^ Young VR, Marchini JS. Mechanisms and nutritional significance of metabolic responses to altered intakes of protein and amino acids, with reference to nutritional adaptation in humans. *Am J Clin Nutr*. (1990)
5. ^ Elango R, Ball RO, Pencharz PB. Indicator amino acid oxidation: concept and application. *J Nutr*. (2008)
6. ^ Humayun MA, et al. Reevaluation of the protein requirement in young men with the indicator amino acid oxidation technique. *Am J Clin Nutr*. (2007)
7. ^ a b c Rafii M, et al. Dietary Protein Requirement of Men >65 Years Old Determined by the Indicator Amino Acid Oxidation Technique Is Higher than the Current Estimated Average Requirement. *J Nutr*. (2016)
8. ^ a b c Rafii M, et al. Dietary protein requirement of female adults >65 years determined by the indicator amino acid oxidation technique is higher than current recommendations. *J Nutr*. (2015)
9. ^ a b Tang M, et al. Assessment of protein requirement in octogenarian women with use of the indicator amino acid oxidation technique. *Am J Clin Nutr*. (2014)
10. ^ Bray GA, et al. Effect of dietary protein content on weight gain, energy expenditure, and body composition during overeating: a randomized controlled trial. *JAMA*. (2012)
11. ^ Thomas DT, Erdman KA, Burke LM. American College of Sports Medicine Joint Position Statement. Nutrition and Athletic Performance. *Med Sci Sports Exerc*. (2016)
12. ^ Jäger R, et al. International Society of Sports Nutrition Position Stand: protein and exercise. *J Int Soc Sports Nutr*. (2017)
13. ^ a b Morton RW, et al. A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults. *Br J Sports Med*. (2018)
14. ^ Wooding DJ, et al. Increased Protein Requirements in Female Athletes after Variable-Intensity Exercise. *Med Sci Sports Exerc*. (2017)
15. ^ Malowany JM, et al. Protein to Maximize Whole-Body Anabolism in Resistance-trained Females after Exercise. *Med Sci Sports Exerc*. (2019)
16. ^ Bandegan A, et al. Indicator amino acid oxidation protein requirement estimate in endurance-trained men 24 h postexercise exceeds both the EAR and current athlete guidelines. *Am J Physiol Endocrinol Metab*. (2019)

17. ^ Bandegan A, et al. Indicator Amino Acid-Derived Estimate of Dietary Protein Requirement for Male Bodybuilders on a Nontraining Day Is Several-Fold Greater than the Current Recommended Dietary Allowance. *J Nutr*. (2017)

18. ^ Joshua L Hudson, et al. Protein Intake Greater Than the RDA Differentially Influences Whole-Body Lean Mass Responses to Purposeful Catabolic and Anabolic Stressors: A Systematic Review and Meta-analysis. *Adv Nutr*. (2020)

19. ^ Antonio J, et al. The effects of consuming a high protein diet (4.4 g/kg/d) on body composition in resistance-trained individuals. *J Int Soc Sports Nutr*. (2014)

20. ^ a b Antonio J, et al. A high protein diet (3.4 g/kg/d) combined with a heavy resistance training program improves body composition in healthy trained men and women--a follow-up investigation. *J Int Soc Sports Nutr*. (2015)

21. ^ a b Antonio J, et al. The effects of a high protein diet on indices of health and body composition--a crossover trial in resistance-trained men. *J Int Soc Sports Nutr*. (2016)

22. ^ Mike Spillane, Darryn S Willoughby. Daily Overfeeding From Protein and/or Carbohydrate Supplementation for Eight Weeks in Conjunction With Resistance Training Does Not Improve Body Composition and Muscle Strength or Increase Markers Indicative of Muscle Protein Synthesis and Myogenesis in Resistance-Trained Males. *J Sports Sci Med*. (2016)

23. ^ Campbell et al. Effects of high versus low protein intake on body composition and maximal strength in aspiring female physique athletes engaging in an 8-week resistance-training program. *Int J Sport Nutr Exe*. (2018)

24. ^ Leaf A, Antonio J. The Effects of Overfeeding on Body Composition: The Role of Macronutrient Composition - A Narrative Review. *Int J Exerc Sci*. (2017)

25. ^ Amy J Hector, Stuart M Phillips. Protein Recommendations for Weight Loss in Elite Athletes: A Focus on Body Composition and Performance. *Int J Sport Nutr Exerc Metab*. (2018)

26. ^ Oliver C Witard, Ina Garthe, Stuart M Phillips. Dietary Protein for Training Adaptation and Body Composition Manipulation in Track and Field Athletes. *Int J Sport Nutr Exerc Metab*. (2019)

27. ^ Helms ER, et al. A systematic review of dietary protein during caloric restriction in resistance trained lean athletes: a case for higher intakes. *Int J Sport Nutr Exerc Metab*. (2014)

28. ^ Aragon AA, et al. International society of sports nutrition position stand: diets and body composition. *J Int Soc Sports Nutr*. (2017)

29. ^ Helms ER, Aragon AA, Fitschen PJ. Evidence-based recommendations for natural bodybuilding contest preparation: nutrition and supplementation. *J Int Soc Sports Nutr*. (2014)

30. ^ Krieger JW, et al. Effects of variation in protein and carbohydrate intake on body mass and composition during energy restriction: a meta-regression 1. *Am J Clin Nutr*.

(2006)

31. ^ Wycherley TP, et al. Effects of energy-restricted high-protein, low-fat compared with standard-protein, low-fat diets: a meta-analysis of randomized controlled trials. *Am J Clin Nutr*. (2012)
32. ^ Kim JE, et al. Effects of dietary protein intake on body composition changes after weight loss in older adults: a systematic review and meta-analysis. *Nutr Rev*. (2016)
33. ^ Mathus-Vliegen EM, Obesity Management Task Force of the European Association for the Study of Obesity. Prevalence, pathophysiology, health consequences and treatment options of obesity in the elderly: a guideline. *Obes Facts*. (2012)
34. ^ Jensen MD, et al. 2013 AHA/ACC/TOS guideline for the management of overweight and obesity in adults: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines and The Obesity Society. *J Am Coll Cardiol*. (2014)
35. ^ Blüher M. Adipose tissue inflammation: a cause or consequence of obesity-related insulin resistance?. *Clin Sci (Lond)*. (2016)
36. ^ Santesso N, et al. Effects of higher- versus lower-protein diets on health outcomes: a systematic review and meta-analysis. *Eur J Clin Nutr*. (2012)
37. ^ Cao L, Morley JE. Sarcopenia Is Recognized as an Independent Condition by an International Classification of Disease, Tenth Revision, Clinical Modification (ICD-10-CM) Code. *J Am Med Dir Assoc*. (2016)
38. ^ Morley JE, et al. Sarcopenia. *J Lab Clin Med*. (2001)
39. ^ Landi F, et al. Sarcopenia as the Biological Substrate of Physical Frailty. *Clin Geriatr Med*. (2015)
40. ^ Kojima G. Frailty as a predictor of disabilities among community-dwelling older people: a systematic review and meta-analysis. *Disabil Rehabil*. (2017)
41. ^ Kojima G. Frailty as a Predictor of Nursing Home Placement Among Community-Dwelling Older Adults: A Systematic Review and Meta-analysis. *J Geriatr Phys Ther*. (2018)
42. ^ Cheng MH, Chang SF. Frailty as a Risk Factor for Falls Among Community Dwelling People: Evidence From a Meta-Analysis. *J Nurs Scholarsh*. (2017)
43. ^ Kojima G. Frailty as a predictor of fractures among community-dwelling older people: A systematic review and meta-analysis. *Bone*. (2016)
44. ^ Kojima G. Frailty as a predictor of hospitalisation among community-dwelling older people: a systematic review and meta-analysis. *J Epidemiol Community Health*. (2016)
45. ^ Brown JC, Harhay MO, Harhay MN. Sarcopenia and mortality among a population-based sample of community-dwelling older adults. *J Cachexia Sarcopenia Muscle*. (2016)

46. ^ Woo T, Yu S, Visvanathan R. Systematic Literature Review on the Relationship Between Biomarkers of Sarcopenia and Quality of Life in Older People. *J Frailty Aging*. (2016)

47. ^ a b Janssen I, Heymsfield SB, Ross R. Low relative skeletal muscle mass (sarcopenia) in older persons is associated with functional impairment and physical disability. *J Am Geriatr Soc*. (2002)

48. ^ Burd NA, Gorissen SH, van Loon LJ. Anabolic resistance of muscle protein synthesis with aging. *Exerc Sport Sci Rev*. (2013)

49. ^ James McKendry, et al. Nutritional Supplements to Support Resistance Exercise in Countering the Sarcopenia of Aging. *Nutrients*. (2020)

50. ^ Mcleod JC, Stokes T, Phillips SM. Resistance Exercise Training as a Primary Countermeasure to Age-Related Chronic Disease. *Front Physiol*. (2019)

51. ^ a b c d Moore DR, et al. Protein ingestion to stimulate myofibrillar protein synthesis requires greater relative protein intakes in healthy older versus younger men. *J Gerontol A Biol Sci Med Sci*. (2015)

52. ^ Coelho-Júnior HJ, et al. Low Protein Intake Is Associated with Frailty in Older Adults: A Systematic Review and Meta-Analysis of Observational Studies. *Nutrients*. (2018)

53. ^ Coelho-Júnior HJ, et al. Relative Protein Intake and Physical Function in Older Adults: A Systematic Review and Meta-Analysis of Observational Studies. *Nutrients*. (2018)

54. ^ Deutz NE, et al. Protein intake and exercise for optimal muscle function with aging: recommendations from the ESPEN Expert Group. *Clin Nutr*. (2014)

55. ^ Bauer J, et al. Evidence-based recommendations for optimal dietary protein intake in older people: a position paper from the PROT-AGE Study Group. *J Am Med Dir Assoc*. (2013)

56. ^ Morley JE, et al. Nutritional recommendations for the management of sarcopenia. *J Am Med Dir Assoc*. (2010)

57. ^ Traylor DA, Gorissen SHM, Phillips SM. Perspective: Protein Requirements and Optimal Intakes in Aging: Are We Ready to Recommend More Than the Recommended Daily Allowance?. *Adv Nutr*. (2018)

58. ^ Mitchell CJ, et al. The effects of dietary protein intake on appendicular lean mass and muscle function in elderly men: a 10-wk randomized controlled trial. *Am J Clin Nutr*. (2017)

59. ^ Nabuco HCG, et al. Effects of Whey Protein Supplementation Pre- or Post-Resistance Training on Muscle Mass, Muscular Strength, and Functional Capacity in Pre-Conditioned Older Women: A Randomized Clinical Trial. *Nutrients*. (2018)

60. ^ Ten Haaf DSM, et al. Protein supplementation improves lean body mass in

physically active older adults: a randomized placebo-controlled trial. *J Cachexia Sarcopenia Muscle*. (2019)

61. ^ Stephens TV, et al. Protein requirements of healthy pregnant women during early and late gestation are higher than current recommendations. *J Nutr*. (2015)
62. ^ Elango R, Ball RO. Protein and Amino Acid Requirements during Pregnancy. *Adv Nutr*. (2016)
63. ^ Imdad A, Bhutta ZA. Maternal nutrition and birth outcomes: effect of balanced protein-energy supplementation. *Paediatr Perinat Epidemiol*. (2012)
64. ^ Dewey KG. Energy and protein requirements during lactation. *Annu Rev Nutr*. (1997)
65. ^ Prentice AM, Goldberg GR, Prentice A. Body mass index and lactation performance. *Eur J Clin Nutr*. (1994)
66. ^ Daly SE, Hartmann PE. Infant demand and milk supply. Part 1: Infant demand and milk production in lactating women. *J Hum Lact*. (1995)
67. ^ Daly SE, Hartmann PE. Infant demand and milk supply. Part 2: The short-term control of milk synthesis in lactating women. *J Hum Lact*. (1995)
68. ^ Motil KJ, et al. Dietary protein and nitrogen balance in lactating and nonlactating women. *Am J Clin Nutr*. (1990)
69. ^ Motil KJ, et al. Whole-body protein turnover in the fed state is reduced in response to dietary protein restriction in lactating women. *Am J Clin Nutr*. (1996)
70. ^ Institute of Medicine. Page 621 in Chapter 10: Protein and Amino Acids. *Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids*. (2005)
71. ^ Institute of Medicine. Page 624 in Chapter 10: Protein and Amino Acids. *Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids*. (2005)
72. ^ Institute of Medicine. Page 630 in Chapter 10: Protein and Amino Acids. *Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids*. (2005)
73. ^ Hay WW, Thureen P. Protein for preterm infants: how much is needed? How much is enough? How much is too much?. *Pediatr Neonatol*. (2010)
74. ^ Fenton TR, et al. Higher versus lower protein intake in formula-fed low birth weight infants. *Cochrane Database Syst Rev*. (2014)
75. ^ Tonkin EL, Collins CT, Miller J. Protein Intake and Growth in Preterm Infants: A Systematic Review. *Glob Pediatr Health*. (2014)
76. ^ Agostoni C, et al. Enteral nutrient supply for preterm infants: commentary from the European Society of Paediatric Gastroenterology, Hepatology and Nutrition

Committee on Nutrition. *J Pediatr Gastroenterol Nutr*. (2010)

77. ^ Arslanoglu S, et al. Fortification of Human Milk for Preterm Infants: Update and Recommendations of the European Milk Bank Association (EMBA) Working Group on Human Milk Fortification. *Front Pediatr*. (2019)
78. ^ Martin CR, Ling PR, Blackburn GL. Review of Infant Feeding: Key Features of Breast Milk and Infant Formula. *Nutrients*. (2016)
79. ^ Gale C, et al. Effect of breastfeeding compared with formula feeding on infant body composition: a systematic review and meta-analysis. *Am J Clin Nutr*. (2012)
80. ^ Oddy WH, et al. Early infant feeding and adiposity risk: from infancy to adulthood. *Ann Nutr Metab*. (2014)
81. ^ Oddy WH. Infant feeding and obesity risk in the child. *Breastfeed Rev*. (2012)
82. ^ Bartok CJ, Ventura AK. Mechanisms underlying the association between breastfeeding and obesity. *Int J Pediatr Obes*. (2009)
83. ^ Liotto N, et al. Clinical evaluation of two different protein content formulas fed to full-term healthy infants: a randomized controlled trial. *BMC Pediatr*. (2018)
84. ^ Oropeza-Ceja LG, et al. Lower Protein Intake Supports Normal Growth of Full-Term Infants Fed Formula: A Randomized Controlled Trial. *Nutrients*. (2018)
85. ^ Tang M. Protein Intake during the First Two Years of Life and Its Association with Growth and Risk of Overweight. *Int J Environ Res Public Health*. (2018)
86. ^ Tang M, Krebs NF. High protein intake from meat as complementary food increases growth but not adiposity in breastfed infants: a randomized trial. *Am J Clin Nutr*. (2014)
87. ^ Tang M, Hendricks AE, Krebs NF. A meat- or dairy-based complementary diet leads to distinct growth patterns in formula-fed infants: a randomized controlled trial. *Am J Clin Nutr*. (2018)
88. ^ a b Tang M, et al. Different Growth Patterns Persist at 24 Months of Age in Formula-Fed Infants Randomized to Consume a Meat- or Dairy-Based Complementary Diet from 5 to 12 Months of Age. *J Pediatr*. (2019)
89. ^ Ahluwalia N, et al. Usual nutrient intakes of US infants and toddlers generally meet or exceed Dietary Reference Intakes: findings from NHANES 2009-2012. *Am J Clin Nutr*. (2016)
90. ^ Elango R, et al. Protein requirement of healthy school-age children determined by the indicator amino acid oxidation method. *Am J Clin Nutr*. (2011)
91. ^ Rodriguez NR. Optimal quantity and composition of protein for growing children. *J Am Coll Nutr*. (2005)
92. ^ Rogerson D. Vegan diets: practical advice for athletes and exercisers. *J Int Soc Sports Nutr*. (2017)

93. ^ a b Moughan P, et al. The assessment of amino acid digestibility in foods for humans and including a collation of published ileal amino acid digestibility data for human foods. *FAO*. (2011)

94. ^ a b . Dietary Protein Quality Evaluation in Human Nutrition. Report of an FAQ Expert Consultation. *FAO Food Nutr Pap*. (2013)

95. ^ Sarwar Gilani G, Wu Xiao C, Cockell KA. Impact of antinutritional factors in food proteins on the digestibility of protein and the bioavailability of amino acids and on protein quality. *Br J Nutr*. (2012)

96. ^ Hou Y, Yin Y, Wu G. Dietary essentiality of "nutritionally non-essential amino acids" for animals and humans. *Exp Biol Med (Maywood)*. (2015)

97. ^ Volpi E, et al. Essential amino acids are primarily responsible for the amino acid stimulation of muscle protein anabolism in healthy elderly adults. *Am J Clin Nutr*. (2003)

98. ^ Wilkinson DJ, et al. Effects of leucine and its metabolite β-hydroxy-β-methylbutyrate on human skeletal muscle protein metabolism. *J Physiol*. (2013)

99. ^ Devries MC, et al. Leucine, Not Total Protein, Content of a Supplement Is the Primary Determinant of Muscle Protein Anabolic Responses in Healthy Older Women. *J Nutr*. (2018)

100. ^ Wolfe RR. Branched-chain amino acids and muscle protein synthesis in humans: myth or reality?. *J Int Soc Sports Nutr*. (2017)

101. ^ van Vliet S, Burd NA, van Loon LJ. The Skeletal Muscle Anabolic Response to Plant- versus Animal-Based Protein Consumption. *J Nutr*. (2015)

102. ^ Yang Y, et al. Myofibrillar protein synthesis following ingestion of soy protein isolate at rest and after resistance exercise in elderly men. *Nutr Metab (Lond)*. (2012)

103. ^ Mitchell CJ, et al. Soy protein ingestion results in less prolonged p70S6 kinase phosphorylation compared to whey protein after resistance exercise in older men. *J Int Soc Sports Nutr*. (2015)

104. ^ Wilkinson SB, et al. Consumption of fluid skim milk promotes greater muscle protein accretion after resistance exercise than does consumption of an isonitrogenous and isoenergetic soy-protein beverage. *Am J Clin Nutr*. (2007)

105. ^ Gran P, et al. Muscle p70S6K phosphorylation in response to soy and dairy rich meals in middle aged men with metabolic syndrome: a randomised crossover trial. *Nutr Metab (Lond)*. (2014)

106. ^ Phillips SM. Nutrient-rich meat proteins in offsetting age-related muscle loss. *Meat Sci*. (2012)

107. ^ Volek JS, et al. Whey protein supplementation during resistance training augments lean body mass. *J Am Coll Nutr*. (2013)

108. ^ Hartman JW, et al. Consumption of fat-free fluid milk after resistance exercise

promotes greater lean mass accretion than does consumption of soy or carbohydrate in young, novice, male weightlifters. *Am J Clin Nutr*. (2007)

109. ^ a b Joy JM, et al. The effects of 8 weeks of whey or rice protein supplementation on body composition and exercise performance. *Nutr J*. (2013)

110. ^ a b Kalman D, et al. Effect of protein source and resistance training on body composition and sex hormones. *J Int Soc Sports Nutr*. (2007)

111. ^ a b Brown EC, et al. Soy versus whey protein bars: effects on exercise training impact on lean body mass and antioxidant status. *Nutr J*. (2004)

112. ^ a b Mobley CB, et al. Effects of Whey, Soy or Leucine Supplementation with 12 Weeks of Resistance Training on Strength, Body Composition, and Skeletal Muscle and Adipose Tissue Histological Attributes in College-Aged Males. *Nutrients*. (2017)

113. ^ Young VR, Pellett PL. Plant proteins in relation to human protein and amino acid nutrition. *Am J Clin Nutr*. (1994)

114. ^ a b Woolf PJ, Fu LL, Basu A. vProtein: identifying optimal amino acid complements from plant-based foods. *PLoS One*. (2011)

115. ^ Churchward-Venne TA, et al. Leucine supplementation of a low-protein mixed macronutrient beverage enhances myofibrillar protein synthesis in young men: a double-blind, randomized trial. *Am J Clin Nutr*. (2014)

116. ^ Norton LE, et al. Leucine content of dietary proteins is a determinant of postprandial skeletal muscle protein synthesis in adult rats. *Nutr Metab (Lond)*. (2012)

117. ^ Brook MS, et al. Skeletal muscle hypertrophy adaptations predominate in the early stages of resistance exercise training, matching deuterium oxide-derived measures of muscle protein synthesis and mechanistic target of rapamycin complex 1 signaling. *FASEB J*. (2015)

118. ^ Damas F, et al. Resistance training-induced changes in integrated myofibrillar protein synthesis are related to hypertrophy only after attenuation of muscle damage. *J Physiol*. (2016)

119. ^ a b Morton RW, McGlory C, Phillips SM. Nutritional interventions to augment resistance training-induced skeletal muscle hypertrophy. *Front Physiol*. (2015)

120. ^ a b Schoenfeld BJ, Aragon AA. How much protein can the body use in a single meal for muscle-building? Implications for daily protein distribution. *J Int Soc Sports Nutr*. (2018)

121. ^ Deutz NE, Wolfe RR. Is there a maximal anabolic response to protein intake with a meal?. *Clin Nutr*. (2013)

122. ^ Kim IY, et al. The anabolic response to a meal containing different amounts of protein is not limited by the maximal stimulation of protein synthesis in healthy young adults. *Am J Physiol Endocrinol Metab*. (2016)

123. ^ Nair KS, Halliday D, Griggs RC. Leucine incorporation into mixed skeletal muscle

protein in humans. *Am J Physiol*. (1988)

124. ^ Ruth M. Leverton. Proteins (chapter 5 of *Food: The Yearbook of Agriculture 1959*). *The United States Department of Agriculture*. (1959)
125. ^ D L Pannemans, D Halliday, K R Westerterp. Whole-body Protein Turnover in Elderly Men and Women: Responses to Two Protein Intakes. *Am J Clin Nutr*. (1995)
126. ^ L. Hambræus. Protein and amino acids in human nutrition. *Elsevier Reference Collection in Biomedical Sciences*. (2014)
127. ^ Moore DR, et al. Ingested protein dose response of muscle and albumin protein synthesis after resistance exercise in young men. *Am J Clin Nutr*. (2009)
128. ^ Symons TB, et al. A moderate serving of high-quality protein maximally stimulates skeletal muscle protein synthesis in young and elderly subjects. *J Am Diet Assoc*. (2009)
129. ^ Aragon AA, Schoenfeld BJ. Nutrient timing revisited: is there a post-exercise anabolic window?. *J Int Soc Sports Nutr*. (2013)
130. ^ Wataru Kume, Jun Yasuda, Takeshi Hashimoto. Acute Effect of the Timing of Resistance Exercise and Nutrient Intake on Muscle Protein Breakdown. *Nutrients*. (2020)